Firmware Build Instructions

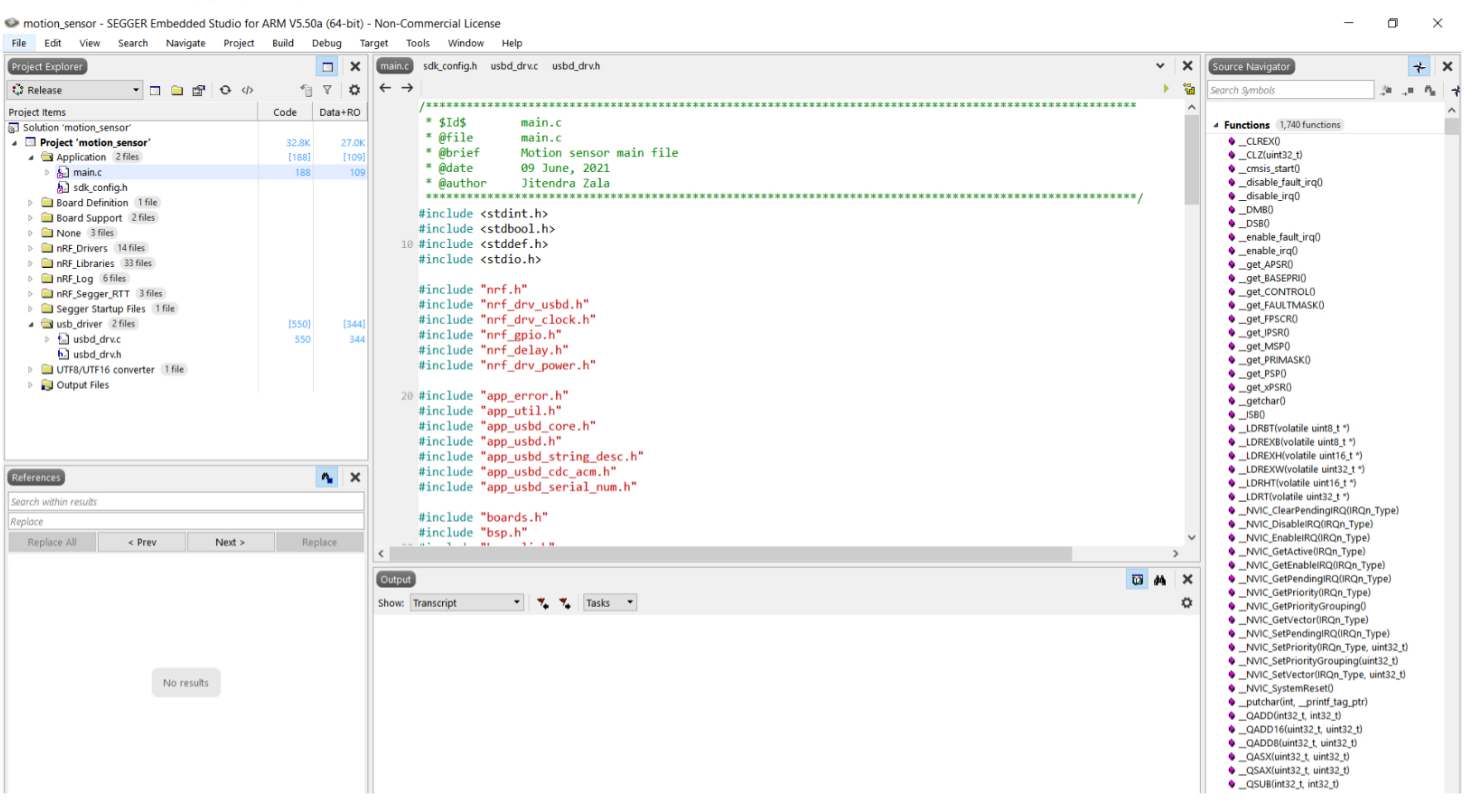
1. Kindly clone main branch of repository from Bitbucket using below link

[https://jitendra_cf@bitbucket.org/jumpcatbucket/jc_dk.git]

2. Go to cloned repository **[jc_dk]**
3. Go to project file as per below reference

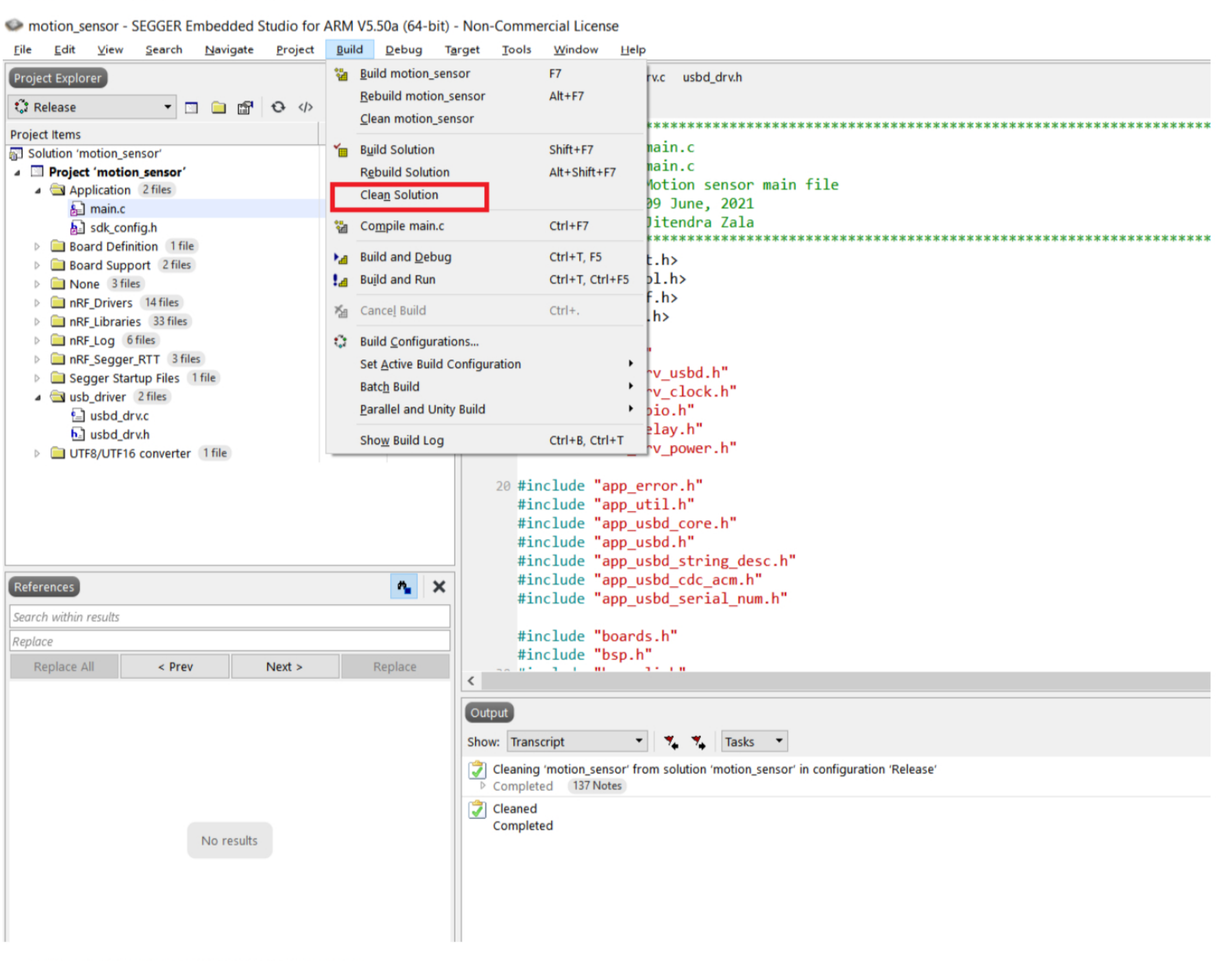
jc_dk\Source_Code\Motion_Sensor\Project_Folder\pca10056\motion_sensor\ses\

4. Open **motion_sensor.emProject** file and project will be opened in Segger Embedded Studio
5. It will ask for license agreement, kindly accept it
6. You can see motion_sensor project opened as per below screenshot.


motion_sensor - SEGGER Embedded Studio for ARM V5.50a (64-bit) - Non-Commercial License
File Edit View Search Navigate Project Build Debug Target Tools Window Help
Project Explorer
Release
Project Items Code Data+RO
Solution 'motion_sensor'
Project 'motion_sensor' 32.8K 27.0K
Application 2 files [188] [109]
main.c 188 109
sdk_config.h
Board Definition 1 file
Board Support 2 files
None 3 files
nRF_Drivers 14 files
nRF_Libraries 33 files
nRF_Log 6 files
nRF_Segger_RTT 3 files
Segger Startup Files 1 file
usb_driver 2 files [550] [344]
usbd_drv.c 550 344
usbd_drv.h
UTF8/UTF16 converter 1 file
Output Files
main.c sdk_config.h usbd_drv.c usbd_drv.h
References
Search within results
Replace
Replace All < Prev Next > Replace
No results
Output
Show: Transcript Tasks
Source Navigator
Search Symbols
Functions 1,740 functions

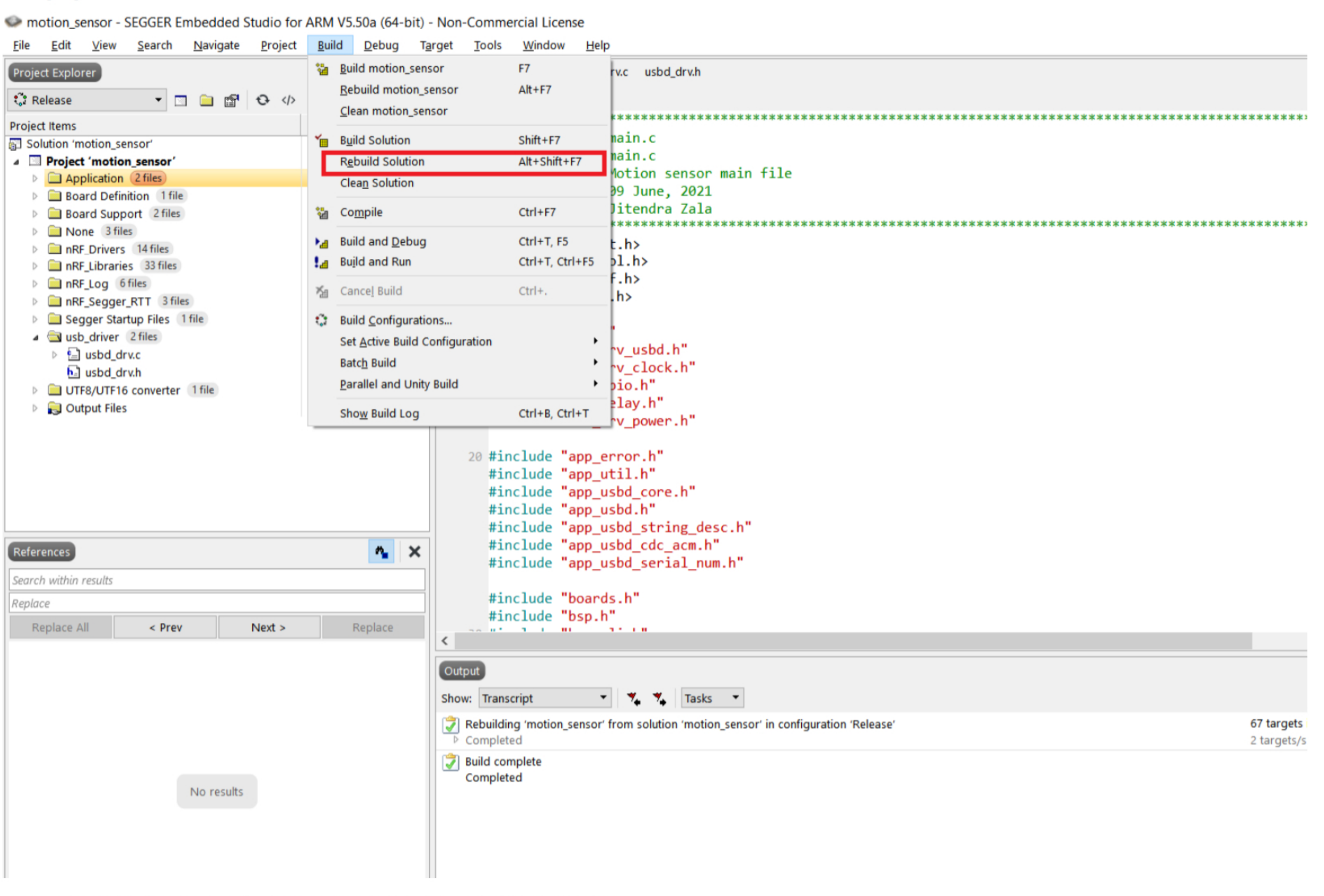
```
/*****************************************************************************
 * $Id$        main.c
 * @file       main.c
 * @brief      Motion sensor main file
 * @date       09 June, 2021
 * @author     Jitendra Zala
 *****************************************************************************/
#include <stdint.h>
#include <stdbool.h>
#include <stddef.h>
#include <stdio.h>

#include "nrf.h"
#include "nrf_drv_usbd.h"
#include "nrf_drv_clock.h"
#include "nrf_gpio.h"
#include "nrf_delay.h"
#include "nrf_drv_power.h"

#include "app_error.h"
#include "app_util.h"
#include "app_usbd_core.h"
#include "app_usbd.h"
#include "app_usbd_string_desc.h"
#include "app_usbd_cdc_acm.h"
#include "app_usbd_serial_num.h"

#include "boards.h"
#include "bsp.h"
```

7. Go to **Build** Menu and Perform **Clean Solution**



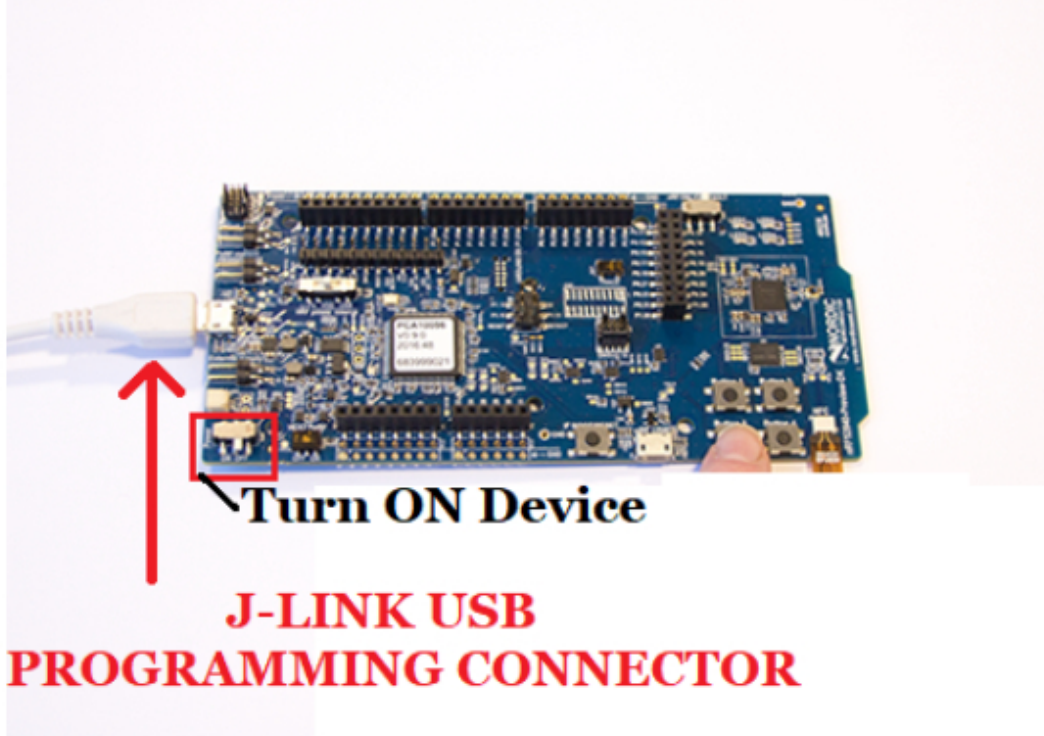
Cleaning 'motion_sensor' from solution 'motion_sensor' in configuration 'Release'
Completed 137 Notes
Cleaned
Completed

8. Again go to **Build** Menu and Perform **Rebuild Solution**

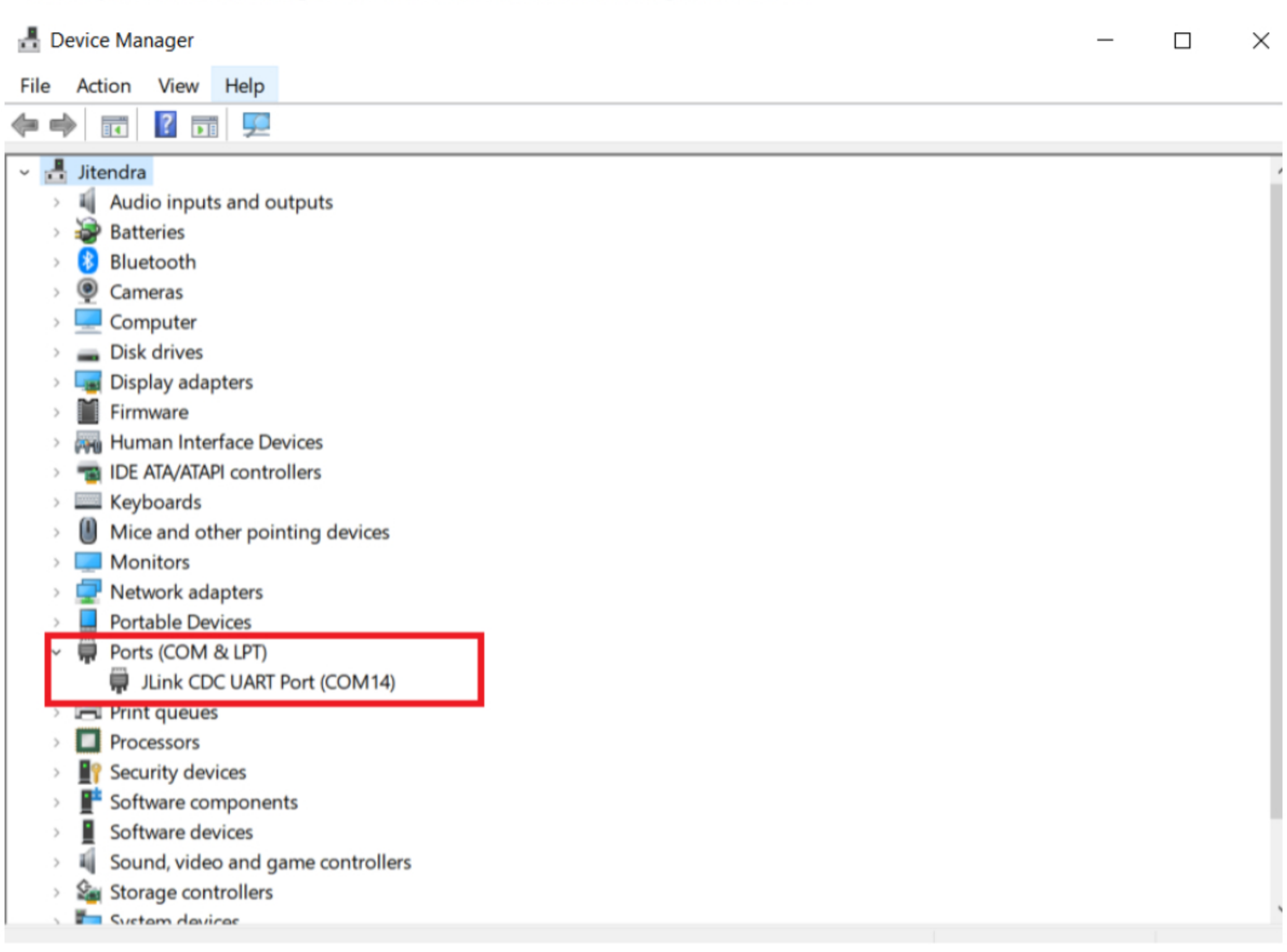

Build motion_sensor F7
Rebuild motion_sensor Alt+F7
Clean motion_sensor
Build Solution Shift+F7
Rebuild Solution Alt+Shift+F7
Clean Solution
Compile Ctrl+F7
Build and Debug Ctrl+T, F5
Build and Run Ctrl+T, Ctrl+F5
Cancel Build Ctrl+.
Build Configurations...
Set Active Build Configuration
Batch Build
Parallel and Unity Build
Show Build Log Ctrl+B, Ctrl+T

Rebuilding 'motion_sensor' from solution 'motion_sensor' in configuration 'Release' 67 targets
Completed 2 targets/s
Build complete
Completed

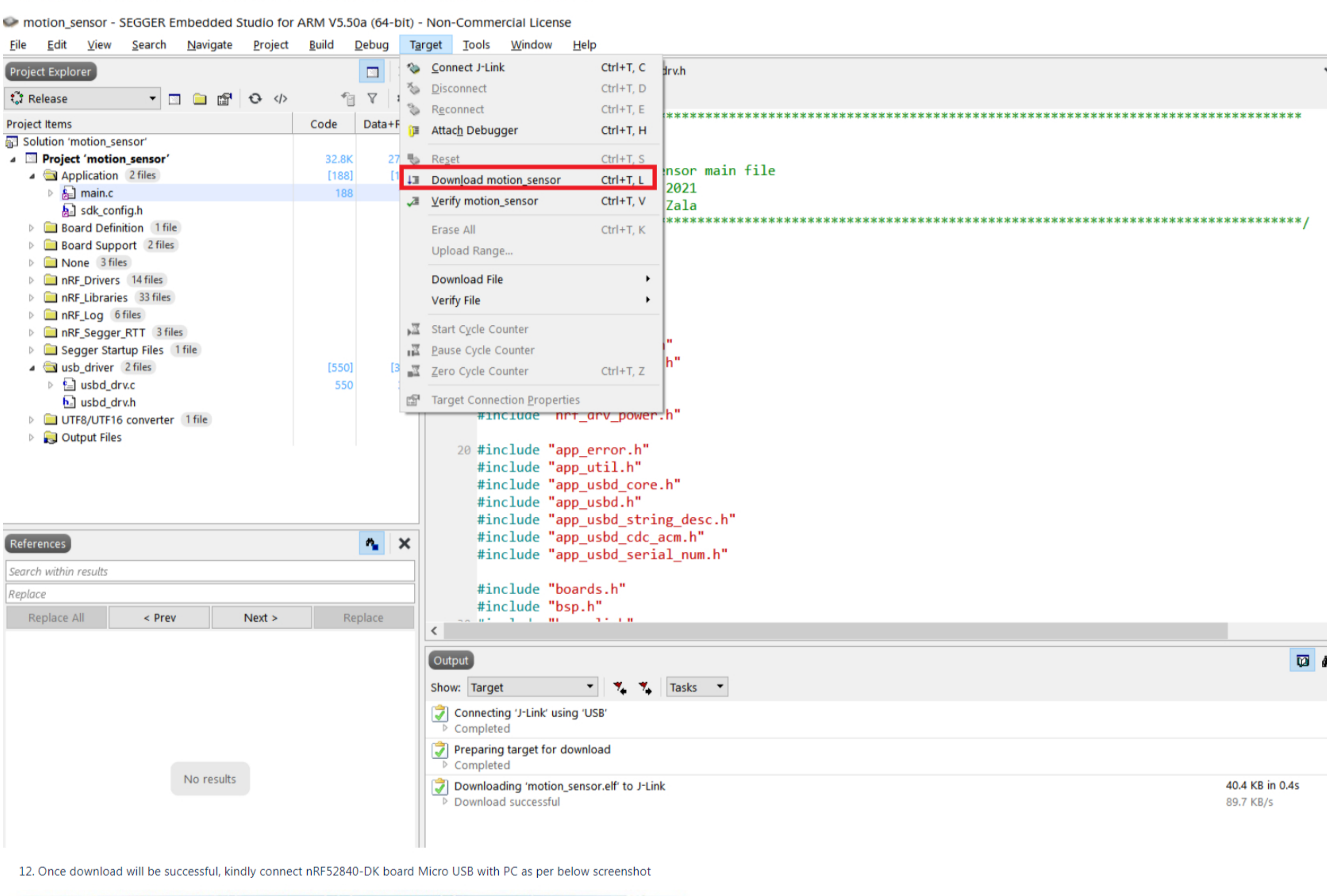
9. Connect **nRF52840-DK* ***board with PC using Micro USB cable and*** ***Turn ON** device as per screenshot.



10. Kindly check in **Device manager** that **J-Link CDC UART Port(COMXX)** is generated or not.

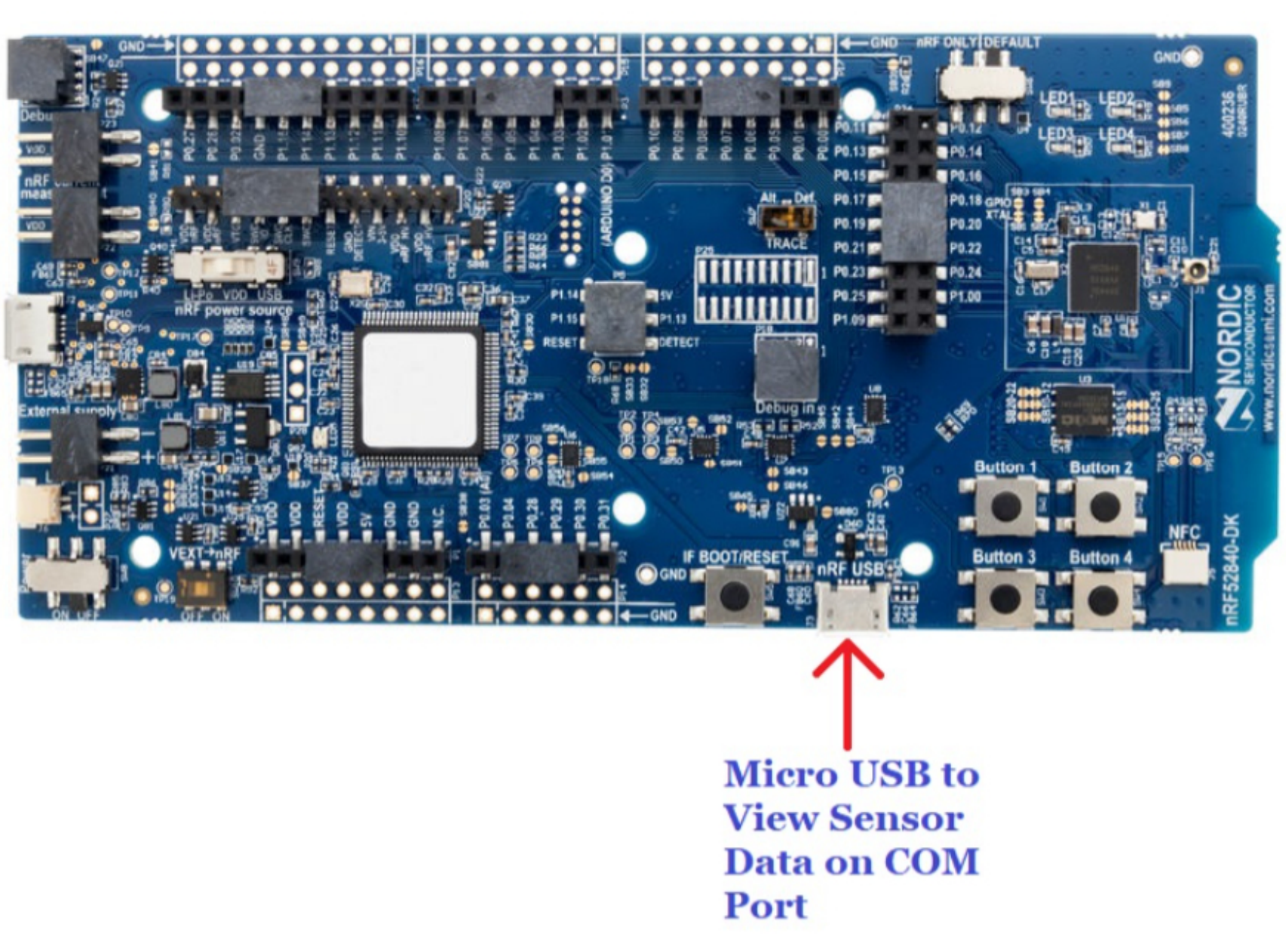


11. Now go back to Segger Embedded Studio, Go to **Target** menu and perform **Download motion_sensor**

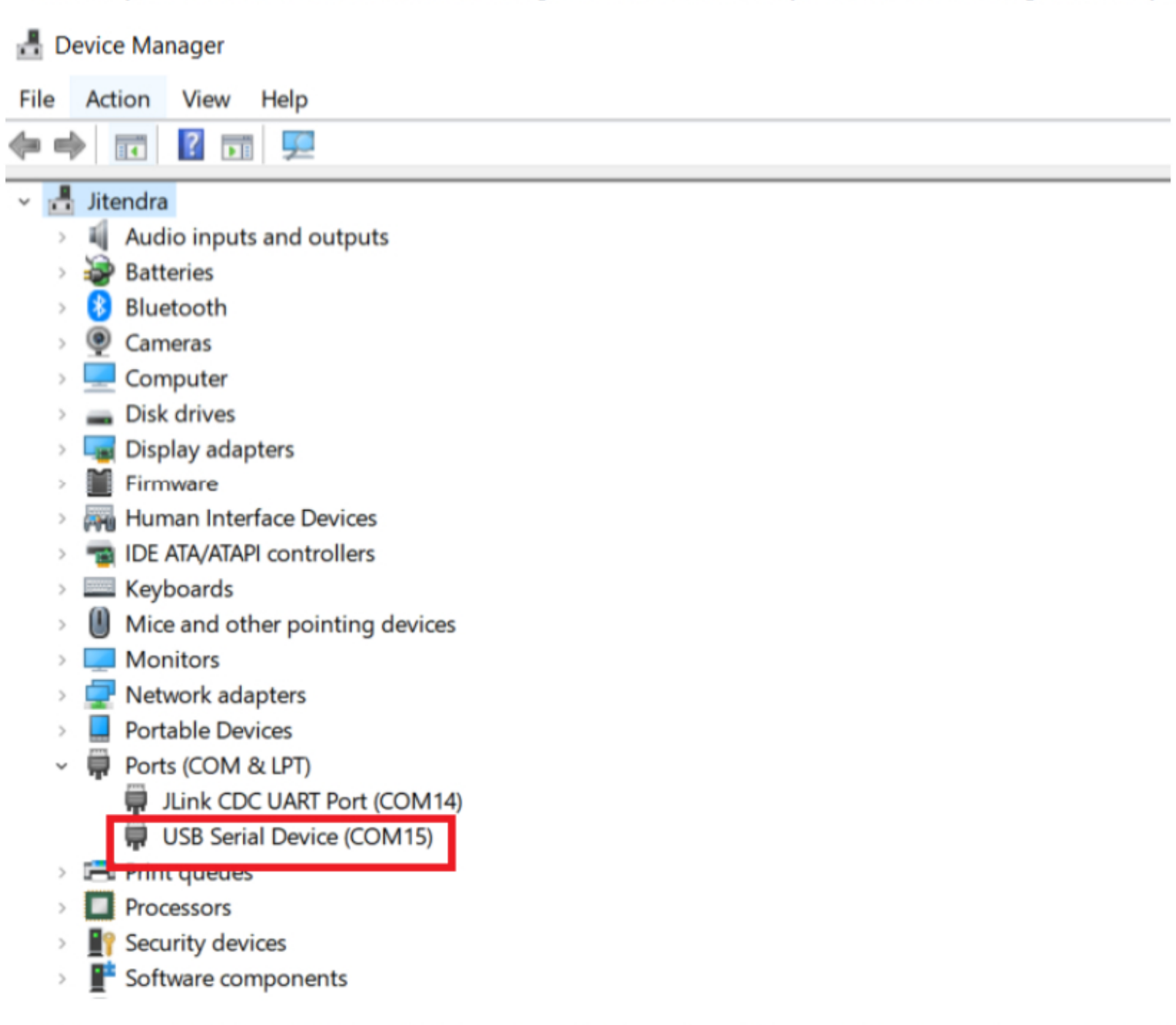


Connecting 'J-Link' using 'USB'
Completed
Preparing target for download
Completed
Downloading 'motion_sensor.elf' to J-Link 40.4 KB in 0.4s
Download successful 89.7 KB/s

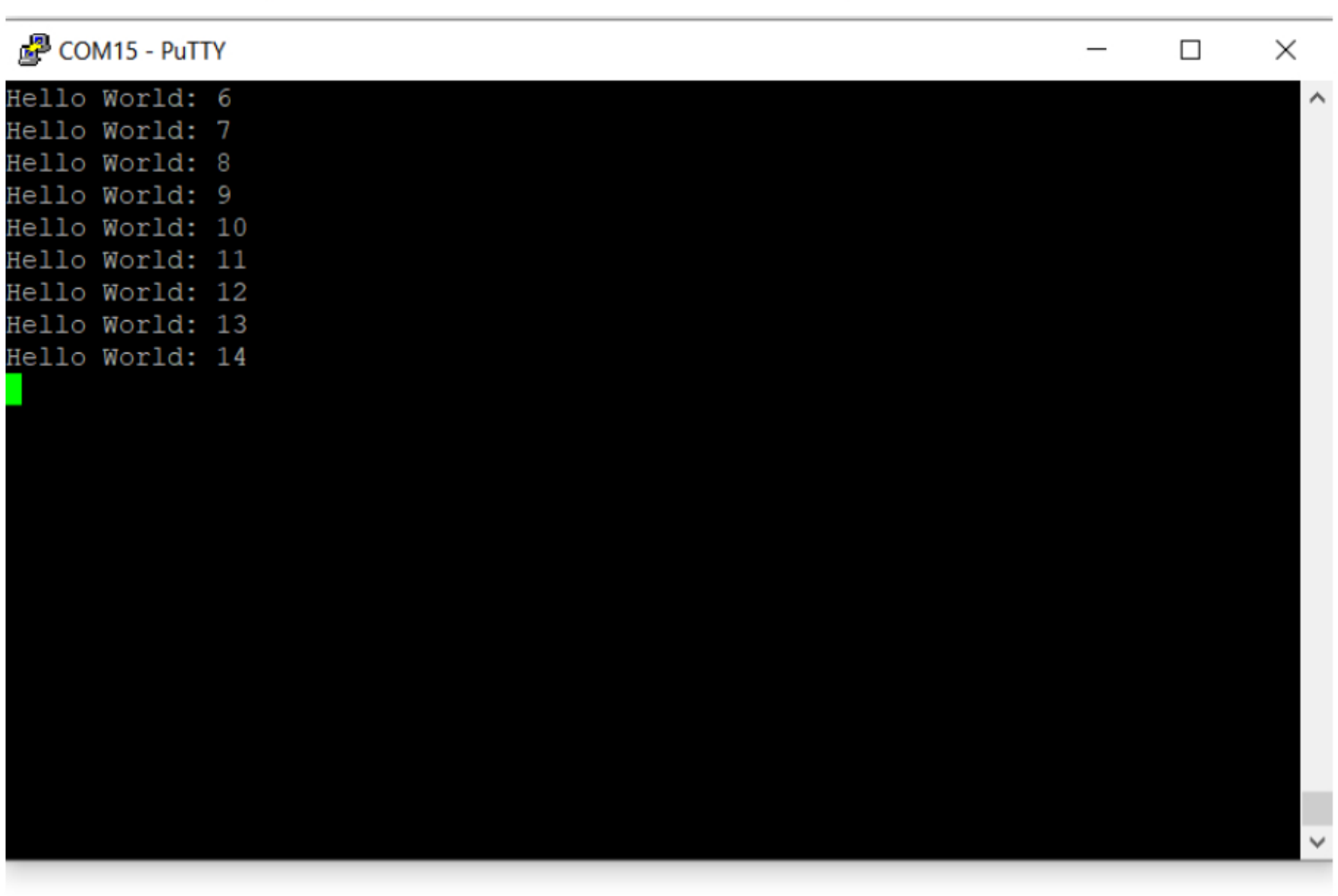
12. Once download will be successful, kindly connect nRF52840-DK board Micro USB with PC as per below screenshot



13. Once you will connect Micro USB with PC, it will generate **COM Port**. Kindly check in **Device manager** and verify **COM port Number**



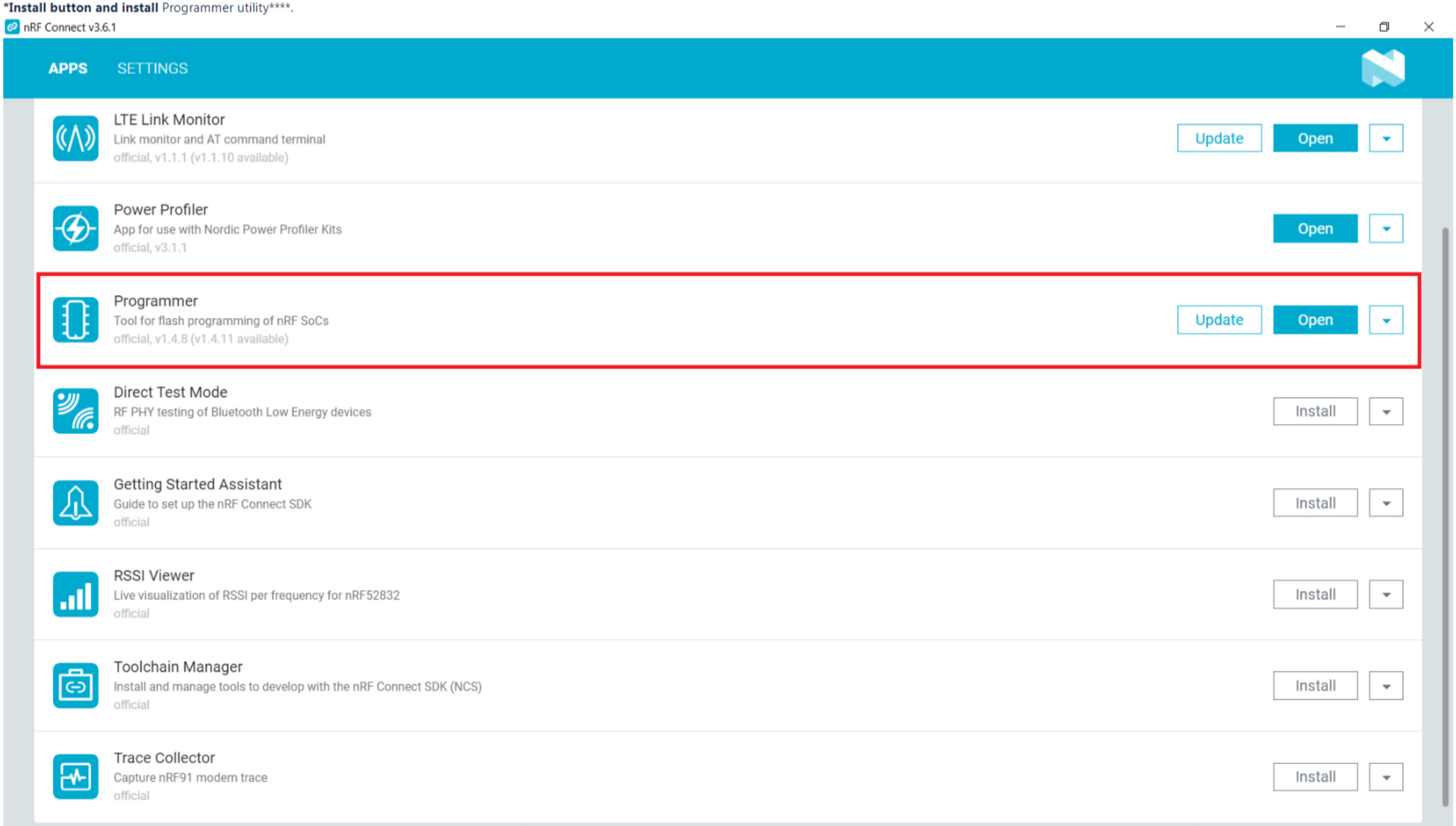
14. Open any **Serial Port/COM Port Utility** like **Putty [Kindly go through https://bitbucket.org/jumpcatbucket/jc_dk/wiki/environment_setup to setup Putty]** , Set COM Port as per Device manager and set **Baudrate 115200**. Once Port will be opened successfully, you will get **Hello World Prints with Count Number** as per below screenshot.



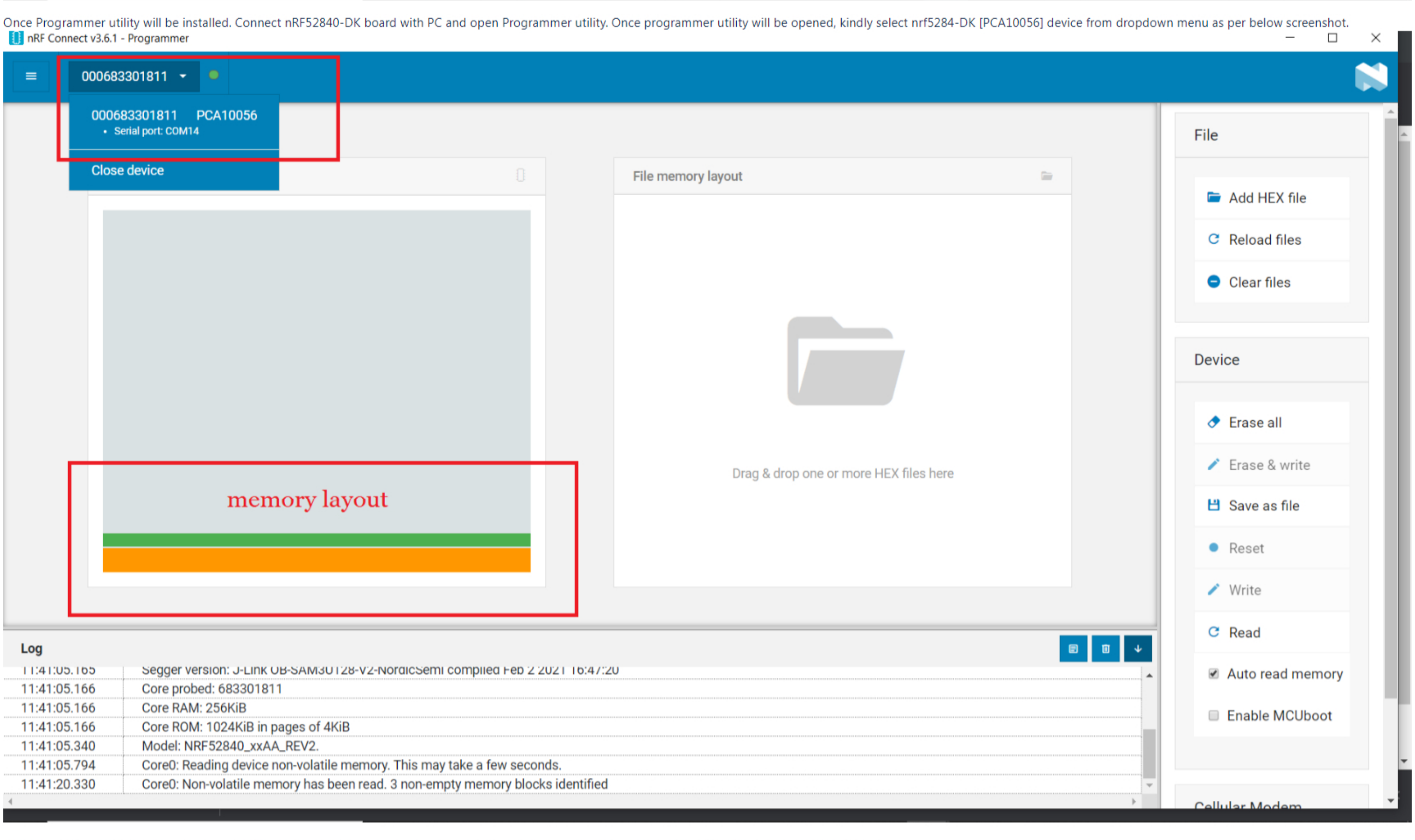
```
Hello World: 6
Hello World: 7
Hello World: 8
Hello World: 9
Hello World: 10
Hello World: 11
Hello World: 12
Hello World: 13
Hello World: 14
```

nRF52840-DK Board Erase Instructions

1. Donwload nRFConnect application from below download link. https://www.nordicsemi.com/-/media/Software-and-other-downloads/Desktop-software/nRF-Connect-for-Desktop/3-6-1/nrfconnectsetup361ia32.exe
2. Install nRFConnect app in Windows PC
3. Once installation will be completed successfully, kindly open **nRFConnect** App. It will show built in application like LTE Link Monitor, Power Profiler, Programmer etc. We need programmer utility as of now. If you see **Install*** ***button, kindly click on*** ***Install button and install** Programmer utility****.

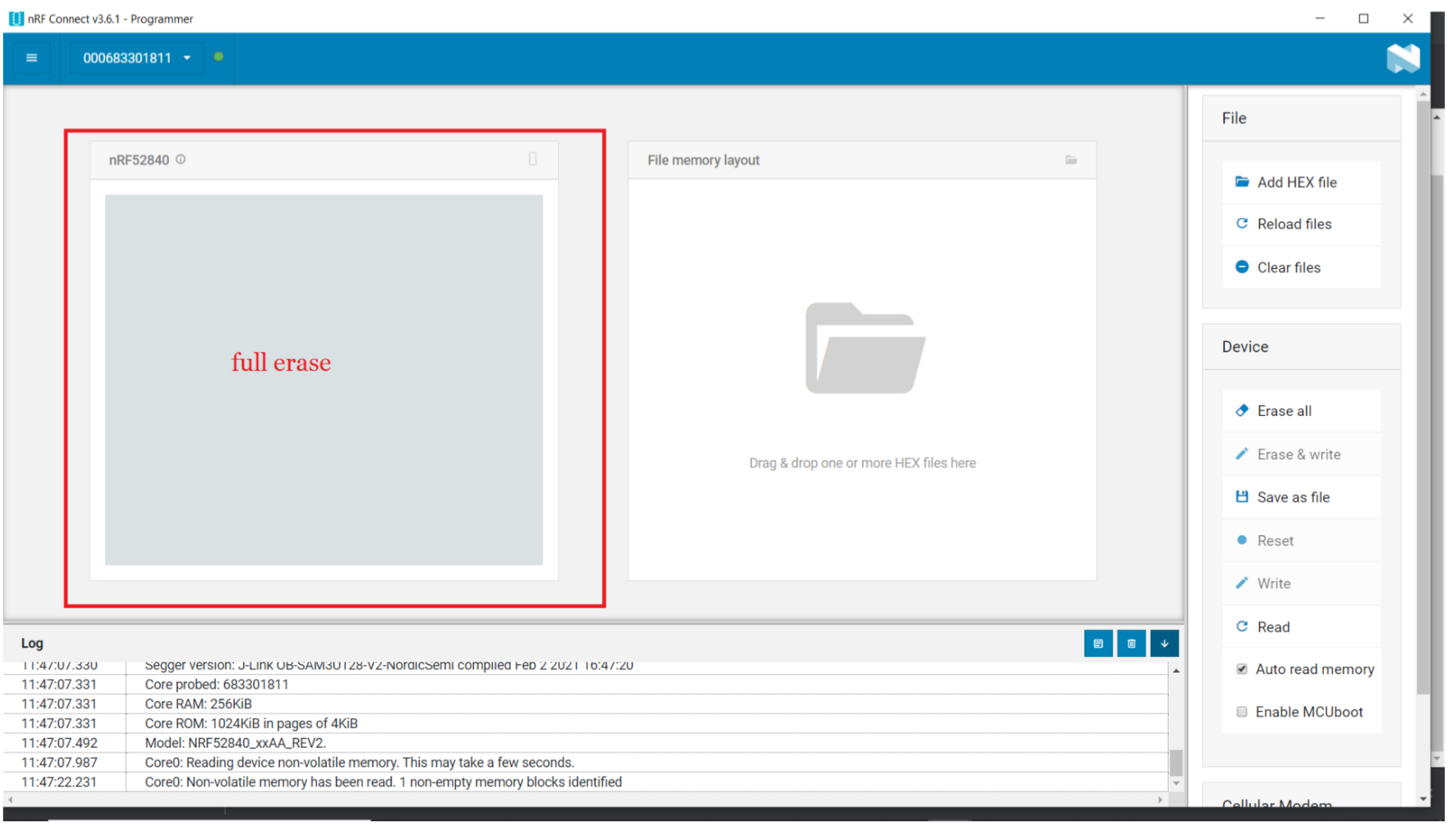

nRF Connect v3.6.1
APPS
SETTINGS
LTE Link Monitor
Link monitor and AT command terminal
official, v1.1.1 (v1.1.10 available)
Update
Open
Power Profiler
App for use with Nordic Power Profiler Kits
official, v3.1.1
Open
Programmer
Tool for flash programming of nRF SoCs
official, v1.4.8 (v1.4.11 available)
Update
Open
Direct Test Mode
RF PHY testing of Bluetooth Low Energy devices
official
Install
Getting Started Assistant
Guide to set up the nRF Connect SDK
official
Install
RSSI Viewer
Live visualization of RSSI per frequency for nRF52832
official
Install
Toolchain Manager
Install and manage tools to develop with the nRF Connect SDK (NCS)
official
Install
Trace Collector
Capture nRF91 modem trace
official
Install

4. Once Programmer utility will be installed. Connect nRF52840-DK board with PC and open Programmer utility. Once programmer utility will be opened, kindly select nrf5284-DK [PCA10056] device from dropdown menu as per below screenshot.



```
11:41:05.165   Segger version: J-Link OB-SAM3U128-V2-NordicSemi compiled Feb 2 2021 16:47:20
11:41:05.166   Core probed: 683301811
11:41:05.166   Core RAM: 256KiB
11:41:05.166   Core ROM: 1024KiB in pages of 4KiB
11:41:05.340   Model: NRF52840_xxAA_REV2.
11:41:05.794   Core0: Reading device non-volatile memory. This may take a few seconds.
11:41:20.330   Core0: Non-volatile memory has been read. 3 non-empty memory blocks identified
```

5. If device will be detected without any error, you will see memory layout as per above screenshot. Now Click on **Erase all** option to perform full erase of device.



```
11:47:07.330   Segger version: J-Link OB-SAM3U128-V2-NordicSemi compiled Feb 2 2021 16:47:20
11:47:07.331   Core probed: 683301811
11:47:07.331   Core RAM: 256KiB
11:47:07.331   Core ROM: 1024KiB in pages of 4KiB
11:47:07.492   Model: NRF52840_xxAA_REV2.
11:47:07.987   Core0: Reading device non-volatile memory. This may take a few seconds.
11:47:22.231   Core0: Non-volatile memory has been read. 1 non-empty memory blocks identified
```

6. Now you have nRF52840-DK board without any firmware or soft device.